Modelling mass TB and TBI screening in South Tarawa (Kiribati) — Technical appendix

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Contents

1 Overview of the Model and Analysis

We developed a deterministic, age-structured compartmental model of *Mycobacterium tuberculosis* (M.tb) transmission, infection, disease, detection, treatment, and prevention. The model uses discrete age groups and continuous time, and is formulated as a system of ordinary differential equations (ODEs). All compartments are stratified by age group and by screening reachability, so that each model state is indexed by both dimensions.

Compartment notation (per age group and reachability stratum). We denote the following compartments:

- **S**: M.tb-*naïve* (never infected).
- **E**: Incipient infection (post-infection with risk of disease progression).
- **C**: Contained infection (immune control without sterilisation - no immediate disease progression risk).
- **X**: Cleared infection (post infection clearance - spontaneous or TPT-induced).
- $\mathbf{D_{SL}}$: Active TB, Subclinical, Low infectiousness.
- $\mathbf{D_{CL}}$: Active TB, Clinical, Low infectiousness.
- $\mathbf{D_{SH}}$: Active TB, Subclinical, High infectiousness.
- $\mathbf{D_{CH}}$: Active TB, Clinical, High infectiousness.
- **T**: On TB treatment.
- **R**: Recovered (post self-recovery or treatment).

We use the term "M.tb naïve" rather than the commonly used "Susceptible" to describe individuals who have never been infected with M.tb, because several non-naïve compartments remain (partially) susceptible to reinfection.

Model overview. Non-disease *M. tuberculosis* infection is modelled through two stages: incipient (E) and contained (C). The contained stage represents a state in which the infection is controlled by the immune system, with no immediate risk of progression to active disease. Breakdown of containment reflects loss of immune control, returning individuals to the incipient stage (E) with a renewed risk of disease progression. Containment can also result in complete clearance of infection, represented by state X.

Progression from E results in subclinical active disease (more or less infectious) that may gain/lose infectiousness and progress/regress between clinical and subclinical states. More details about the disease categorisation and transitions are provided in Section 4. Only *clinical* forms of disease are detected through passive case finding; while active screening may potentially detect any form of TB. Screening programmes add time-limited detection flows for both TB disease and non-disease M. tb infection states (Section 7).

Age-specific demography and mixing are implemented explicitly to capture the effects of population structure and contact patterns on transmission dynamics (Sections 3–5.2).

The model is additionally stratified by screening reachability, distinguishing individuals who can participate in active screening from those who are hard to reach. The latter receive no active screening and are assumed to have greater susceptibility to infection and poorer access to passive case detection (Section 6).

Simulation period and initial conditions. The model is simulated from 1850 to 2035. The initial population size is set to the earliest available WPP estimate for the modelled population (1950), and no population growth is applied before that first data year, so the population size remains constant over the initial period and follows the observed trajectory thereafter (Section 3). The epidemic is seeded at the start of the simulation with 100 individuals in the clinical, high-infectiousness disease state (D_{CH}), with the remainder of the population placed in the M.tb-naïve compartment (S). The long period preceding the first calibration data allows the age distribution and the TB epidemic to reach an endemic equilibrium well before the period informed by observations, so that results are insensitive to the size and composition of the initial seed.

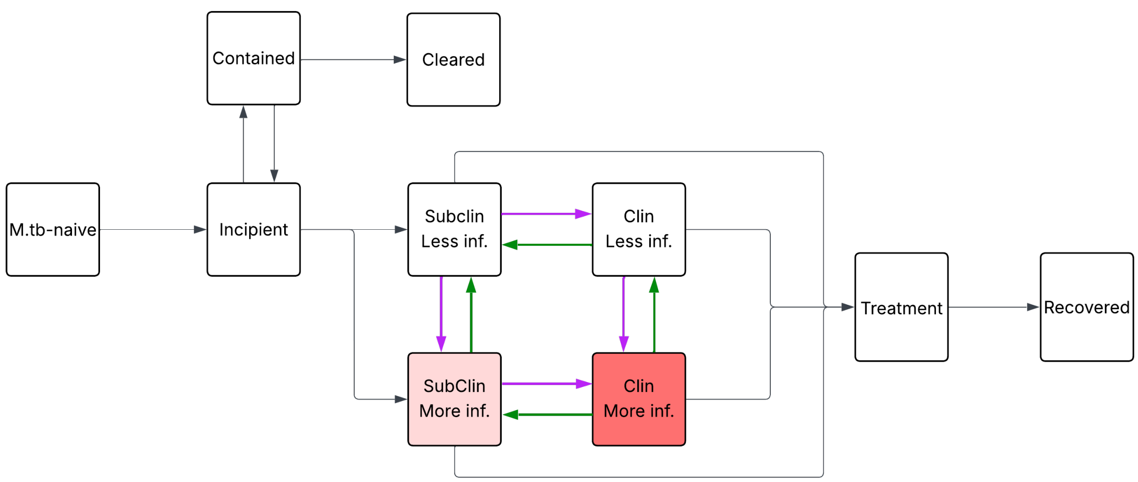

Figure 1: Schematic of model compartments and transitions.

Model structure.

Parameter treatment in calibration. Parameters with specified prior distributions are estimated in the Bayesian calibration while those without priors are fixed. Estimated classes include core transmission (baseline transmission rate and population-scaling exponent), social mixing (background mixing level, age assortativity spread, and parent–child mixing strength), selected early-infection and disease-transition rates (clearance, immunity breakdown, clinical progression, gain of infectiousness), the passive-detection scale-up profile, and a subset of diagnostic sensitivities.

Model use in analysis. The model is used in two stages: (i) to reconstruct the historical epidemic trajectory by calibrating uncertain parameters to epidemiological and intervention data; and (ii) to project forward under alternative intervention scenarios (screening and preventive treatment). Parameter estimation is performed in a Bayesian framework with Differential Evolution Metropolis algorithm (DEMetropolisZ); posterior samples are propagated to quantify uncertainty in projected outcomes.

2 Software and Code

The model is implemented in Python using `summer2`, enabling symbolic specification and efficient numerical ODE generation. Computations are accelerated via `jax` just-in-time compilation, which lowers the ODE system to optimised machine code after first execution; subsequent simulations require ∼200 ms each, enabling large-scale Bayesian calibration.

All model code, calibration routines, analysis scripts, and interactive notebooks are available at `https://github.com/monash-emu/kiribati_tb_modelling`.

The repository includes instructions for installation and reproduction of results (Need to work on this). The project environment is managed using `pixi` for reproducible dependency management and straightforward installation.

3 Demographics

Each age and disease state in the model is subject to the appropriate all-cause and TB-specific mortality rates. Non-TB deaths are calculated using time-variant and age-specific mortality rates, and births are injected into the youngest M.tb-naïve compartment to reproduce the observed population size over time. This approach allows the model to match observed population trajectories while maintaining internal demographic dynamics. Age-specific mortality rates and population size over time are taken from the United Nations World Population Prospects (WPP) data. We implement ageing as first-order transitions between adjacent age groups, with rates calculated as the inverse of the age group's span and applied to all age strata except the oldest one.

4 Natural History of Infection and Disease

We represent early TB infection dynamics using two non-disease states: an *incipient* state (E) and a *contained* state (C). Individuals newly infected with *M. tb* enter E, where infection is unstable and at risk of progression. From E, they may either progress to subclinical active TB or undergo immune-mediated containment, moving to C. Contained infection may subsequently clear entirely to X (uninfected/cleared) or experience immunological failure, returning to the incipient state E. Active TB is stratified by both clinical presentation (subclinical vs. clinical) and infectiousness (low vs. high), yielding four disease compartments: $D_{SL}, D_{CL}, D_{SH}, D_{CH}$. Individuals with subclinical disease may progress to clinical disease, and clinical disease may regress to subclinical; similarly, infectiousness may increase (low → high) or decrease (high → low). However, transitions are restricted to a single dimension at a time. Individuals may change clinical status (subclinical ↔ clinical) or infectiousness (low ↔ high), but simultaneous transitions in both dimensions are not allowed. Self-recovery is allowed only from subclinical states, with recovered individuals entering R. In contrast, TB-induced mortality rates are only applied to the clinical states. Treatment is available for all active states, with successful treatment leading to recovery (R) and unsuccessful treatment resulting in relapse to subclinical low infectiousness (D_{SL}) or death.

Early infection. Progression from E is age-specific (three categories: < 5, 5–14, 15+) and partitioned into low vs high infectiousness at first presentation. Containment rates are also age-specific. Clearance and immune breakdown are age-invariant.

Clinical and infectiousness transitions. Movements between subclinical and clinical states, and between low and high infectiousness, govern within-disease dynamics. The cross-sectional observations available from the completed PEARL screening phase constrain the relative sizes of the four disease compartments, but not the speed at which individuals move between them: the same distribution of prevalent TB across states can be reproduced by slow or by rapid transitions in both directions.

To resolve this identifiability problem, the two regression rates (clinical → subclinical and high → low infectiousness) are fixed at a common value, while the two corresponding progression rates (subclinical → clinical and low → high infectiousness) are estimated during calibration. In the base case the common regression rate is set to 1 per year, corresponding to an average of one year spent in a given state before regression. Under this constraint, the estimated progression rates adjust so that the model reproduces the observed distribution of prevalent TB across the four states.

Because this common value is an assumption rather than an estimate, it is varied in sensitivity analysis across 0.5, 1, 2 and 3 per year, with the model recalibrated at each value. This yields configurations that all reproduce the observed cross-sectional distribution of disease states while differing markedly in the overall speed of movement across the TB spectrum.

Infectiousness weighting. Let σ_S denote the relative infectiousness of subclinical vs clinical disease, and σ_L the relative infectiousness of low vs high. Define the active-disease index set

$$\mathcal{K} = \{SL, CL, SH, CH\},$$

with relative infectiousness weights w_k given by $w_{CH} = 1$, $w_{CL} = \sigma_L$, $w_{SH} = \sigma_S$, $w_{SL} = \sigma_S\sigma_L$. Children under 15 years contribute zero infectiousness, implemented by setting $w_k = 0$ in those age groups.

4.1 Generalised Transmission and Infectiousness Weights

We interpolate between frequency- and density-dependent transmission using the exponent $\kappa \in [0, 1]$ applied to the contacted population size. Because mixing between reachability strata is homogeneous (Section 6), the force of infection depends on age group only. For age group i it is

$$\lambda_i(t) = \beta_0 \, N_{\text{ref}}^{\kappa} \sum_j \tilde{C}_{ij}(t) \, \frac{I_j^{\text{eff}}(t)}{N_j(t)^{\kappa}},$$

where $\tilde{C}(t)$ is the spectral-radius-normalised age-mixing matrix (Section 5.2). Here N_{ref} denotes the total population in year 2020 and is included purely as a scaling constant for more efficient calibration. This rescaling stabilises the magnitude of the transmission coefficient β_0 across different values of κ, preventing its calibrated value from varying by orders of magnitude as the dependence on population size changes. The effective infectious pool and the population size of age group j are both aggregated over the two reachability strata $s \in \mathcal{S} = \{\text{r}, \text{u}\}$:

$$I_j^{\text{eff}}(t) = \sum_{s \in \mathcal{S}} \sum_{k \in \mathcal{K}} w_k \, D_{k,j,s}(t), \qquad N_j(t) = \sum_{s \in \mathcal{S}} N_{j,s}(t).$$

Infectiousness weights w_k do not differ by reachability stratum; only susceptibility does, through the multiplier θ_s^{sus} introduced in Section 6. Individuals $< 15\,\text{y}$ contribute zero infectiousness (weights set to zero in those age groups).

4.2 Reinfection Pathways

Reinfection is permitted from C, X, and R with relative susceptibilities ρ_C (contained) and $\rho_X = \rho_R$ (cleared and recovered), and a child susceptibility modifier σ^{child} applied when infection arises from S in $< 15\,\text{y}$ groups. This adjustment captures the effect of BCG vaccination, which is typically administered in infancy and provides protection against infection for several years.

5 Age Stratification and Mixing

5.1 Ageing and Age-Specific Adjustments

Age groups. The population is stratified into $G = 8$ age groups with lower bounds

$$\mathcal{A} = \{0, 3, 5, 10, 15, 18, 40, 65\}.$$

The final group is open-ended (65+).

The breaks were chosen so that every age-dependent process and observation used in the analysis aligns exactly with a group boundary, avoiding the need to apportion parameters or targets within groups:

- **3 years** is the minimum age of eligibility for screening under the PEARL algorithm, below which no active screening or preventive treatment is offered.
- **5 and 15 years** delimit the three age categories used for the early-infection parameters (< 5, 5–14, 15+), and 15 years additionally separates the age groups in which susceptibility to infection is reduced to reflect BCG protection and infectiousness is set to zero.
- **10 years** separates children receiving symptom screening only from older individuals additionally receiving chest X-ray and, in the PEARL algorithm, frontline Xpert.
- **18 years** allows the model to reproduce the earlier estimate of TST positivity among adults aged 18 years and over, which is used as a calibration target (Section 9.1).
- **40 and 65 years** subdivide the adult population to capture differences in background mortality and in contact patterns across adult ages.

Ageing and adjustments. Ageing is implemented as first-order transitions between adjacent age groups. Several age-specific adjustments and interventions are applied in the model (see previous sections for details):

- Age-specific early infection rates using three categories: < 5 years, 5–14 years, and 15+ years.
- Reduced susceptibility to infection from S in individuals aged < 15 years to reflect BCG effect.
- Infectiousness set to zero for individuals aged < 15 years.
- Age-specific background mortality.
- Age-specific screening interventions.

5.2 Age-Mixing Matrix

Notation and data sources. Let $i, j \in \{1, \ldots, G\}$ index age groups and $[a_i, b_i]$ denote the single ages in group i. Calendar time is t (years). We use UN World Population Prospects data to inform age-specific population sizes (1950–2100) and age-specific fertility rates (1950–2023). For modelled years outside data ranges, quantities are clamped to the nearest available year.

Overview. At time t, we construct an age-specific contact matrix

$$\mathbf{C}(t) = \big(C_{ij}(t)\big)_{i,j=1}^{G},$$

where $C_{ij}(t)$ is the per-capita rate at which age group i contacts age group j. Construction proceeds by: (1) building a symmetric per-pair intensity matrix $\mathbf{S}(t)$; (2) scaling columns by the contacted group size; (3) normalising by the spectral radius. We combine three components: uniform background mixing, assortative age mixing, and parent–child mixing. Three scalar parameters govern these components (background level, assortativity spread, and parent–child mixing strength) and are estimated.

Population weights within age groups. To allow non-uniform single-age distributions within groups, let $N(a, t)$ be the population at age a and time t. For group i,

$$w_i(a, t) = \frac{N(a, t)}{\sum_{a' \in [a_i, b_i]} N(a', t)}, \qquad \sum_{a \in [a_i, b_i]} w_i(a, t) = 1.$$

Symmetric per-pair intensity. For each i, j,

$$S_{ij}(t) = b + S_{ij}^{\text{assort}}(t) + \kappa_{\text{pc}}\, S_{ij}^{\text{pc}}(t), \quad S_{ij}(t) = S_{ji}(t),$$

where b is the background level (`bg_mixing`), S^{assort} is an exponential kernel with spread ℓ (`a_spread`), and S^{pc} encodes parent–child contacts from fertility patterns, with strength κ_{pc} (`pc_strength`). Smaller values of ℓ correspond to more strongly assortative mixing by age.

Assortative age mixing.

$$S_{ij}^{\text{assort}}(t) = \sum_{a \in [a_i, b_i]} \sum_{c \in [a_j, b_j]} w_i(a, t)\, w_j(c, t)\, \frac{1}{\ell} \exp\!\left(-\frac{|a - c|}{\ell}\right).$$

Parent–child mixing. Let $p_{\text{pc}}(\Delta, y)$ be the probability of a parent–child age gap Δ for a child born in year y. With $a_{\text{child}} = \min(a, c)$ and birth year $t - a_{\text{child}}$,

$$S_{ij}^{\text{pc}}(t) = \sum_{a \in [a_i, b_i]} \sum_{c \in [a_j, b_j]} w_i(a, t)\, w_j(c, t)\, p_{\text{pc}}\big(|a - c|,\ t - a_{\text{child}}\big).$$

Population scaling and asymmetry. The per-pair matrix becomes an asymmetric WAIFW matrix by column scaling with the contacted group size:

$$C_{ij}(t) = S_{ij}(t)\, N_j(t).$$

Normalisation. Let $\rho(t)$ be the spectral radius of $\mathbf{C}(t)$. We use the normalised matrix

$$\tilde{\mathbf{C}}(t) = \frac{\mathbf{C}(t)}{\rho(t)}.$$

6 Stratification by Screening Reachability

Population-wide screening never reaches the entire population, and the individuals who are not reached may differ epidemiologically from those who participate. To capture this, the model is stratified into two mutually exclusive strata: a *screening-reachable* stratum, comprising individuals who can participate in active screening, and a *hard-to-reach* stratum, comprising individuals who do not. This stratification is applied to all compartments and is crossed with the age stratification, so that each model state is indexed by both age group i and reachability stratum $s \in \{\mathrm{r}, \mathrm{u}\}$.

Population split. The proportion of the population in the screening-reachable stratum is denoted π and is fixed at 0.85, based on participation observed during the completed phase of the PEARL study. The initial population is split between the two strata according to π and $1 - \pi$ within every compartment and age group.

The stratum sizes are held at these proportions over time. Because there are no transitions between the reachable and unreachable strata, this is achieved by (i) splitting the additional births used to reproduce population growth between the two strata according to π and $1 - \pi$, and (ii) returning all deaths (background, TB-related, and on-treatment) to the M.tb-naïve compartment of the youngest age group *within the same reachability stratum* from which they originated. Reachability is therefore a fixed, lifelong attribute in the model, and the fraction of the population that is hard to reach does not drift as the epidemic or the interventions evolve. Mixing between the reachable and unreachable strata is assumed to be homogeneous. In the absence of data on the contact patterns of individuals not reached by screening, this is a deliberately conservative assumption: assortative mixing within the unreachable stratum would be expected to further concentrate residual transmission in that group.

Epidemiological differences between strata. Barriers to participating in screening are unlikely to occur in isolation and may coincide with living conditions or health-seeking behaviours that also affect TB risk and access to routine diagnosis. Two multiplicative adjustments are therefore applied to the unreachable stratum, relative to the reachable stratum:

- **Susceptibility to infection.** All infection and reinfection flows (from S, C, X and R) are multiplied by θ_{sus} (`rel_sus_unreachable`) in the unreachable stratum. The base-case value is 1.5, i.e. hard-to-reach individuals are assumed to acquire infection at 1.5 times the rate of reachable individuals experiencing the same force of infection. This factor is applied to the susceptibility side of transmission only; the relative infectiousness of individuals with TB does not differ by stratum.
- **Passive detection.** The time-varying passive detection rate $\delta(t)$ (Section 7) is multiplied by θ_{det} (`rel_detection_unreachable`) in the unreachable stratum. The base-case value is 0.5, reflecting an assumption of reduced access to routine diagnostic services in the same population that is difficult to reach through active screening.

Both parameters are fixed rather than estimated, as the available data do not inform them; θ_{sus} is varied in sensitivity analysis.

Access to screening. All active screening flows, for both TB disease and non-disease M.tb infection, are set to zero in the unreachable stratum. Screening programmes therefore act exclusively on the screening-reachable stratum. As a consequence, the maximum achievable population coverage is π: a nominal coverage of 85% of the total population corresponds to screening essentially the whole reachable stratum, whereas lower nominal coverage levels represent incomplete participation among reachable individuals (Section 7).

Calibration and reporting. Because the epidemiological observations collected during the completed PEARL screening phase were made among participants, the corresponding calibration targets (measured TB prevalence and TST positivity) are compared with model outputs restricted to the reachable stratum. Historical notifications, which arise from passive detection in the whole population, are compared with model outputs aggregated across both strata. Projected burden outcomes are likewise reported for the entire modelled population unless otherwise stated, and the share of TB incidence arising from the unreachable stratum is tracked as an additional output.

7 TB Detection, Treatment and Screening

7.1 Detection

Passive detection. Passive case finding applies *only to clinical disease*. Subclinical disease is not detected passively. Denote the time-varying passive detection profile by $\delta(t)$, which follows a smooth hyperbolic-tangent scale-up from an earlier (lower) level to a recent level:

$$\delta(t) = \delta_{\mathrm{rec}} \left[\phi + (1-\phi) \left(\frac{\tanh\big(\sigma\,(t-t^*)\big)}{2} + \frac{1}{2} \right) \right].$$

Here, δ_{rec} is the recent detection level, $\phi \in (0,1]$ is the past-to-recent ratio, $\sigma > 0$ controls steepness, and t^* is the inflection time.

Figure 2 illustrates an example profile for $\delta_{\mathrm{rec}} = 0.7$, $\phi = 0.5$, $\sigma = 0.15$ and $t^* = 2005$.

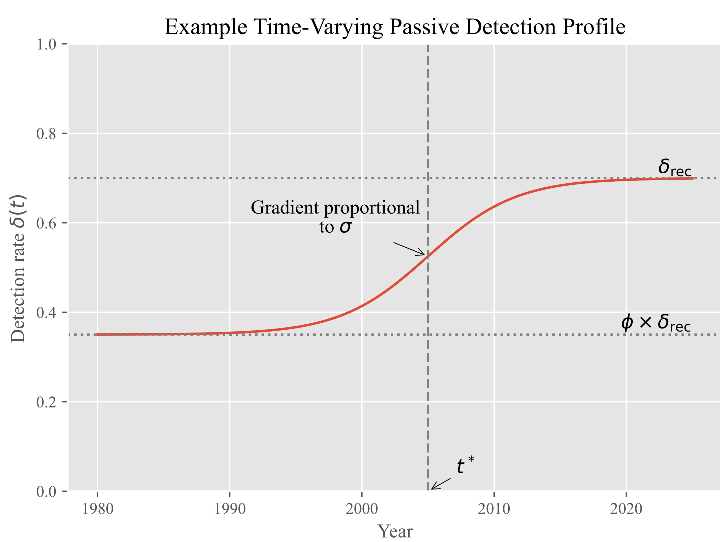

Figure 2: Example time-varying passive detection profile for clinical TB disease.

7.2 Screening programmes

We consider a set of time-limited screening interventions implemented over the year 2026. Each screening programme is defined by a screening algorithm, a target population (by age), and a coverage level. Three coverage levels are explored: 65%, 75% and 85%. These are expressed as proportions of the *total* target population, that is, with hard-to-reach individuals included in the denominator.

Because screening acts only on the reachable stratum (Section 6), which comprises 85% of the population, the maximum coverage that can be achieved is 85%. The highest level considered, 84.5%, therefore corresponds to screening 99.4% of reachable individuals in the eligible age groups and is treated as the practical maximum; it is set marginally below 85% because complete coverage of the reachable stratum would require an infinite screening rate. The two lower levels represent incomplete participation among reachable individuals.

For each programme, the total coverage is translated into a constant per-capita screening rate applied to the reachable stratum over the intervention period. Specifically, if c denotes the target coverage of the total population (expressed as a proportion), π the reachable fraction, and Δt the duration of the campaign, the screening rate is defined as $-\frac{\log(1-c/\pi)}{\Delta t}$.

This ensures that, under a constant hazard assumption, a proportion c/π of the eligible reachable population, and hence a proportion c of the total target population, is screened over the intervention period. The screening rate is implemented as a time-varying function that is zero outside the intervention window and constant during the campaign.

Screening interventions may act on TB infection or active TB disease states. For each screening algorithm, screened individuals have a probability of being correctly identified defined by the sensitivity of the relevant algorithm for that compartment. For TB screening, positively screened individuals are moved to the treatment compartment (T). For TPT, only a fraction of the positively screened individuals transition to the cleared compartment (X) based on the probability of successful TPT completion.

Screening algorithms. Sensitivities in the model are properties of a complete *algorithm* rather than of an individual test. For a given algorithm, the sensitivity assigned to a compartment is the probability that a person occupying that state, and screened under that algorithm, is correctly identified and referred for treatment.

We consider the following approaches for the detection of TB disease, in increasing order of intensity:

- **Symptom screening (SSx)**: screening on the basis of reported symptoms alone. Clinical disease is assumed to be identified with certainty and subclinical disease not at all, so that this algorithm cannot detect subclinical TB by construction.

- **Chest X-ray (CXR)**: symptom screening combined with chest X-ray, with bacteriological confirmation of those with an abnormal result. The added value of CXR lies in the detection of subclinical disease, and the corresponding sensitivities are estimated during calibration (separately for less- and more-infectious subclinical TB), informed by the prevalence observed during the completed PEARL screening phase.

- **PEARL**: symptom screening and CXR, with Xpert MTB/RIF Ultra additionally used as a frontline test in a subset of participants. Operational constraints during implementation meant that frontline Xpert was applied to 35% of those screened, so the algorithm is represented as a mixture of the two branches:

$$s_k^{\text{PEARL}} = 0.35\, s_k^{\text{Xpert}} + 0.65\, s_k^{\text{CXR}}, \qquad k \in \mathcal{K},$$

where s_k^{Xpert} is the sensitivity of the branch that adds frontline Xpert, fixed at one for all four disease states.

By construction, $s_k^{\text{SSx}} \leq s_k^{\text{CXR}} \leq s_k^{\text{Xpert}}$ for every disease state k.

Screening for non-disease M.tb infection is performed using the **tuberculin skin test (TST)**, applied to the incipient and contained states with a common probability of a positive result. Individuals identified in this way are offered TPT, and only those completing it transition to the cleared state.

Age targeting. Screening programmes are targeted to specific age groups by applying multiplicative coverage adjustments during model age stratification. Age groups not targeted by a given programme have zero screening rate.

Scenario structure. We consider three screening algorithms:

- **PEARL:** the algorithm implemented during the completed phase of the PEARL study. Individuals aged 10 years and above receive CXR, with Xpert additionally used as a frontline test in 35% of them, reflecting operational constraints on its implementation. Children aged 3–9 years receive symptom screening, and TST-guided TPT is offered to all individuals aged 3 years and above.
- **CXR-TST:** Xpert is removed from frontline screening. CXR is used for individuals aged 10 years and above, with symptom screening for ages 3–9 years and TST-guided TPT for ages 3 years and above.
- **Disease screening only:** both Xpert and TST-guided TPT are removed, and screening is restricted to CXR among individuals aged 10 years and above. No screening is performed in children under 10 years, and no preventive treatment is provided.

Each algorithm is combined with each of the three coverage levels, giving a total of nine screening scenarios. These are compared with a baseline scenario in which no active screening is implemented. This framework allows us to assess the incremental population-level contribution of frontline Xpert, of TST-guided preventive treatment, and of increasing screening coverage.

7.3 Treatment outcomes

During treatment, individuals may experience three mutually exclusive outcomes: successful recovery, relapse, or death. Recovery occurs at a rate depending on the time-varying treatment success rate:

$$\text{Recovery:} \quad \text{rate} = \frac{\text{TSR}(t)}{\tau},$$

where $\text{TSR}(t)$ denotes the proportion of successful treatments at time t, and τ is the average treatment duration.

Age-specific relapse and treatment-death rates are calculated to account for both background mortality and the fixed proportion of treatment episodes that result in negative outcomes. Let μ_i

be the natural mortality rate for age group i. The fixed fraction of negative outcomes resulting in death is denoted f_{neg}.

The expected proportion of individuals dying from natural causes while on treatment is:

$$p_i^{\text{nat\_death}} = 1 - e^{-\tau \mu_i}.$$

The target proportion of deaths due to treatment failure (excluding natural mortality) is calculated as the fraction of negative outcomes allocated to death:

$$p^{\text{tx\_death}} = (1 - \text{TSR}(t)) \cdot f_{\text{neg}}.$$

To ensure that the overall probability of death on treatment remains non-negative after accounting for natural mortality, the treatment-death rate is defined as:

$$\mu_i^{\text{TX}} = \frac{\max\{p^{\text{tx\_death}} - p_i^{\text{nat\_death}}, 0\}}{\tau}.$$

Finally, the age-specific relapse rate is derived from the residual fraction of unsuccessful treatment episodes, after subtracting both treatment deaths and natural deaths:

$$\phi_i = \frac{1 - \text{TSR}(t) - \tau \mu_i^{\text{TX}} - p_i^{\text{nat\_death}}}{\tau}.$$

These equations ensure that the sum of recovery, relapse, and treatment-death probabilities over the course of treatment equals one, while explicitly incorporating time-varying treatment success and age-specific background mortality.

8 Ordinary Differential Equations

8.1 Notations

We present the ODEs for a generic age group $i \in \{1, \ldots, G\}$ and reachability stratum $s \in \mathcal{S} = \{\text{r}, \text{u}\}$, where r denotes screening-reachable and u hard-to-reach individuals (unreachable by screening, Section 6). Every compartment therefore carries both indices, e.g. $S_{i,s}$. No flow connects the two reachability strata, so s is a fixed attribute of an individual and the strata are coupled only through the shared force of infection.

Let $\alpha_i(t)$ be the ageing rate from group i to $i+1$ ($\alpha_G = 0$), and $\mu_i(t)$ the all-cause mortality rate; neither depends on s. For any compartment variable $U \in \{S, E, C, X, D_{SL}, D_{SH}, D_{CL}, D_{CH}, T, R\}$, define the (within-stratum) ageing operator

$$\mathcal{A}_i[U_{\cdot,s}] \equiv -\alpha_i\, U_{i,s} + \alpha_{i-1}\, U_{i-1,s}, \qquad \alpha_0 = 0.$$

Let $N_{i,s}(t)$ be the total population of age group i in stratum s, and $N_i(t) = \sum_s N_{i,s}(t)$. The force of infection $\lambda_i(t)$ is given above and is common to both strata.

Reachability multipliers. The three multiplicative adjustments that distinguish the two strata (Section 6) are written as stratum-indexed factors:

$$\theta_s^{\text{sus}} = \begin{cases} 1, & s = \text{r}, \\ \theta_{\text{sus}}, & s = \text{u}, \end{cases} \qquad \theta_s^{\text{det}} = \begin{cases} 1, & s = \text{r}, \\ \theta_{\text{det}}, & s = \text{u}, \end{cases} \qquad \theta_s^{\text{scr}} = \begin{cases} 1, & s = \text{r}, \\ 0, & s = \text{u}, \end{cases}$$

applied respectively to all infection and reinfection flows, to the passive detection rate, and to all active screening flows. All other natural-history, treatment and demographic parameters are shared between strata.

Entries into the model. Let $\mathcal{B}(t)$ be the additional entry rate used to reproduce the observed population growth, allocated between strata according to the fixed reachable fraction π, with $\pi_{\mathrm{r}} = \pi$ and $\pi_{\mathrm{u}} = 1-\pi$. All deaths are recycled into the M.tb-naïve compartment of the youngest age group of the stratum in which they occurred, giving the stratum-specific re-entry term

$$\mathcal{D}_s(t) = \sum_{i=1}^{G} \left[\mu_i N_{i,s} + \sum_{k\in\mathcal{K}} \mu_k^{TB}\, D_{k,i,s} + \mu_i^{TX}\, T_{i,s} \right].$$

Together, these two devices keep the total population on its observed trajectory while holding the reachable and hard-to-reach fractions constant at π and $1-\pi$.

Early infection parameters (age-specific). For each age group i, let γ_i be the containment rate from E to C, ϵ_i the total progression rate from E to subclinical disease. The proportion progressing initially to "high infectiousness subclinical" rather than "low infectiousness subclinical" is denoted by p. Then progression risks are $\epsilon_i(1-p)$ to D_{SL} and $\epsilon_i p$ to D_{SH}. From C, let ξ be the clearance rate and ω the breakdown rate.

Clinical/infectiousness transitions and recovery. Let χ^+ (subclinical→clinical) and χ^- (clinical→subclinical) be clinical transitions; σ^+ (low→high) and σ^- (high→low) be infectiousness transitions. For notational convenience, we define χ^k to denote χ^+ when k corresponds to a subclinical state and χ^- when k corresponds to a clinical state; analogously, σ^k denotes σ^+ for low infectiousness states and σ^- for high infectiousness states.

Let $\mathcal{K} = \{SL, SH, CL, CH\}$ index the active disease states, where S and C denote subclinical and clinical disease, respectively, and L and H denote low and high infectiousness. Thus, for example, SL represents subclinical disease with low infectiousness. We introduce the notation $\bar{k}$ to denote the state obtained by switching the clinical status of k (e.g., $\overline{SL} = CL$) and $\tilde{k}$ to denote the state obtained by switching the infectiousness status of k (e.g., $\tilde{SL} = SH$).

Self-recovery from subclinical states occurs at rate ρ_{SC}.

Screening and passive detection hazards. Let $\zeta_{k,i}(t)$ denote the (time- and age-specific) *screening* detection hazard applied to disease state k in the reachable stratum. Screening is unavailable to the hard-to-reach stratum and passive detection is reduced there, while passive detection applies only to *clinical* disease. The total detection hazard is therefore

$$\eta_{k,i,s}^{\mathrm{tot}}(t) = \begin{cases} \theta_s^{\mathrm{scr}}\, \zeta_{k,i}(t), & k \in \{SL, SH\} \quad \text{(subclinical; no passive detection)},\\ \theta_s^{\mathrm{scr}}\, \zeta_{k,i}(t) + \theta_s^{\mathrm{det}}\, \delta(t), & k \in \{CL, CH\} \quad \text{(clinical; screening + passive)}. \end{cases}$$

Since $\theta_{\mathrm{u}}^{\mathrm{scr}} = 0$, this reduces to $\eta_{k,i,\mathrm{u}}^{\mathrm{tot}} = 0$ for subclinical states and to $\theta_{\mathrm{det}}\, \delta(t)$ for clinical states in the hard-to-reach stratum. The rate of TPT in the reachable stratum is denoted $u_i^{\mathrm{TPT}}(t)$ and combines the TST screening rate and the preventive treatment completion rate; in the hard-to-reach stratum it is set to zero, so that

$$u_{i,s}^{\mathrm{TPT}}(t) = \theta_s^{\mathrm{scr}}\, u_i^{\mathrm{TPT}}(t).$$

State-specific auxiliary terms. The following terms depend on the disease state and, where indicated, on age, but not on the reachability stratum. Subclinical-only self-recovery by state:

$$r_k^{\mathrm{SC}} = \begin{cases} \rho_{SC}, & k \in \{SL, SH\},\\ 0, & k \in \{CL, CH\}. \end{cases}$$

TB-specific mortality by active state (applied to clinical states only, and depending on infectiousness):

$$\mu_k^{TB} = \begin{cases} 0, & k \in \{SL, SH\}, \\ \mu_{CL}^{TB}, & k = CL, \\ \mu_{CH}^{TB}, & k = CH. \end{cases}$$

Age-specific progression seeding from E_i into active disease:

$$b_{k,i} = \begin{cases} \epsilon_i(1-p), & k = SL, \\ \epsilon_i p, & k = SH, \\ 0, & k \in \{CL, CH\}. \end{cases}$$

8.2 System of ODEs

$\forall i \in \{1, \ldots, G\}$ and $\forall s \in \mathcal{S} = \{\mathrm{r}, \mathrm{u}\}$:

$$\frac{dS_{i,s}}{dt} = \mathbf{1}_{\{i=1\}} \left(\pi_s \, \mathcal{B}(t) + \mathcal{D}_s(t)\right) - \theta_s^{\text{sus}} \, \sigma_i^{\text{child}} \, \lambda_i S_{i,s} - \mu_i S_{i,s} + \mathcal{A}_i[S_{\cdot,s}], \tag{1}$$

$$\frac{dE_{i,s}}{dt} = \theta_s^{\text{sus}} \, \lambda_i \left[\sigma_i^{\text{child}} S_{i,s} + \rho_C C_{i,s} + \rho_X (X_{i,s} + R_{i,s})\right] + \omega \, C_{i,s} - \left(\gamma_i + \epsilon_i + \mu_i + u_{i,s}^{\text{TPT}}\right) E_{i,s} + \mathcal{A}_i[E_{\cdot,s}], \tag{2}$$

$$\frac{dC_{i,s}}{dt} = \gamma_i E_{i,s} - \left(\omega + \xi + \mu_i + u_{i,s}^{\text{TPT}}\right) C_{i,s} + \mathcal{A}_i[C_{\cdot,s}], \tag{3}$$

$$\frac{dX_{i,s}}{dt} = \xi \, C_{i,s} + u_{i,s}^{\text{TPT}} \, (E_{i,s} + C_{i,s}) - \theta_s^{\text{sus}} \, \rho_X \lambda_i X_{i,s} - \mu_i X_{i,s} + \mathcal{A}_i[X_{\cdot,s}], \tag{4}$$

$$\begin{aligned} \frac{dD_{k,i,s}}{dt} = \; & b_{k,i} \, E_{i,s} + \underbrace{\chi^{\bar{k}} D_{\bar{k},i,s} - \chi^{k} D_{k,i,s}}_{\text{clinical transitions}} + \underbrace{\sigma^{\tilde{k}} D_{\tilde{k},i,s} - \sigma^{k} D_{k,i,s}}_{\text{infectiousness transitions}} + \underbrace{\mathbf{1}_{\{k=SL\}} \phi_i T_{i,s}}_{\text{treatment relapse}} \\ & - \left(\eta_{k,i,s}^{\text{tot}}(t) + \mu_i + \mu_k^{TB} + r_k^{\text{SC}}\right) D_{k,i,s} + \mathcal{A}_i[D_{k,\cdot,s}], \qquad k \in \mathcal{K}, \end{aligned} \tag{5}$$

$$\frac{dT_{i,s}}{dt} = \sum_{k \in \mathcal{K}} \eta_{k,i,s}^{\text{tot}}(t) \, D_{k,i,s} - \left(\frac{\text{TSR}(t)}{\tau} + \phi_i + \mu_i^{TX} + \mu_i\right) T_{i,s} + \mathcal{A}_i[T_{\cdot,s}], \tag{6}$$

$$\frac{dR_{i,s}}{dt} = \sum_{k \in \mathcal{K}} r_k^{\text{SC}} D_{k,i,s} + \frac{\text{TSR}(t)}{\tau} \, T_{i,s} - \theta_s^{\text{sus}} \, \rho_X \lambda_i R_{i,s} - \mu_i R_{i,s} + \mathcal{A}_i[R_{\cdot,s}]. \tag{7}$$

9 Model Calibration

9.1 Data Sources

TB notifications. Historical TB notifications were obtained from the Kiribati National Tuberculosis Programme, recorded by locality within South Tarawa. Because the model represents the part of South Tarawa remaining to be screened by PEARL in January 2026, the notification series was restricted to the localities east of, and excluding, Betio, Bairiki and Nanikai. A proportion of notifications each year had no locality recorded; these were allocated to the modelled area in proportion to its share of the South Tarawa population (63.3%), giving the imputed annual series used for calibration.

Treatment success rate. The proportion of TB treatment episodes ending in success, TSR(t), was specified as a time-variant input rather than estimated. Annual reported values for Kiribati were used for 1995–2022, together with a recent value for 2025 and two earlier anchor points representing the era before effective chemotherapy (0% in 1945) and its early introduction (50% in 1950). Values are linearly interpolated between these points and held constant outside the range.

The treatment success rate determines the recovery rate on treatment and, together with the fixed fraction of unsuccessful episodes ending in death, the relapse and treatment-death rates described in Section 7.

Demographic data. Population size by single year of age (1950–2100), age-specific deaths, and age-specific fertility rates (1950–2023) were taken from the United Nations World Population Prospects, as described in Sections 3 and 5.2.

External contact matrix. No empirical contact survey data are available for Kiribati. To constrain the age-mixing matrix estimated within the model (Section 5.2), we generated a synthetic contact matrix using the `conmat` R package. A contact model was fitted to the POLYMOD contact survey data and the corresponding survey populations, and then used to predict all-setting contact rates for the age distribution of the modelled South Tarawa population in 2025, using the same age breaks as the model. The resulting matrix was normalised by its spectral radius so that it is comparable with the model's normalised mixing matrix, and enters the likelihood through the matrix-distance term described below.

Epidemiological observations from PEARL. Measured TB prevalence (using the PEARL algorithm and using CXR), the proportions of prevalent TB that were subclinical and more infectious, and age-specific TST positivity were obtained from the completed phase of PEARL screening. Because these observations were made among screening participants, they are compared with model outputs restricted to the reachable stratum (Section 6). An additional, earlier estimate of TST positivity among adults aged 18 years and over was taken from a previous survey conducted in Kiribati.

9.2 Calibration Targets and Likelihood

Calibration targets include TB notifications, TST positivity by age, prevalence by disease form, fraction subclinical, fraction high infectiousness, and a penalty on the distance between the modelled mixing matrix and the external matrix described in Section 9.1. For prevalence targets, model outputs incorporate test sensitivity parameters. Each target k consists of one or more observations $y_{k,t}$ made at times t, each compared with the corresponding model output $m_{k,t}(\theta)$ through a normal likelihood:

$$y_{k,t} \sim \mathcal{N}\big(m_{k,t}(\theta), \sigma_k^2\big).$$

A single standard deviation is used for all observations belonging to a given target, defined relative to the mean of the observed series $\bar{y}_k$:

$$\sigma_k = \frac{\mathrm{tol}_k}{1.96}\,\bar{y}_k, \qquad \mathrm{tol}_k = \begin{cases} 0.40, & k = \text{notifications}, \\ 0.20, & \text{otherwise}, \end{cases}$$

so that the 95% interval of the likelihood spans ±20% (or ±40% for notifications) of the mean observed value. The total log-likelihood is

$$\log L(\theta) = \sum_k \sum_t \Big(-\frac{(y_{k,t} - m_{k,t}(\theta))^2}{2\sigma_k^2} - \frac{1}{2}\log(2\pi\sigma_k^2)\Big).$$

Mixing matrix penalty. The estimated mixing matrix is not compared with the external synthetic matrix element by element. Instead, the discrepancy between the two is summarised by a single scalar, the Canberra distance between the spectral-radius-normalised model matrix $\tilde{\mathbf{C}}(t)$ evaluated in 2025 and the normalised external matrix $\mathbf{C}^{\text{ext}}$:

$$d(\theta) = \sum_{i,j} \frac{\left|\tilde{C}_{ij} - C_{ij}^{\text{ext}}\right|}{\left|\tilde{C}_{ij}\right| + \left|C_{ij}^{\text{ext}}\right| + \varepsilon},$$

with $\varepsilon = 10^{-10}$ included to avoid division by zero. This relative, element-wise normalisation gives comparable weight to discrepancies in low- and high-contact cells, rather than allowing the largest matrix entries to dominate.

The distance is treated as an additional target with an observed value of zero, $0 \sim \mathcal{N}\big(d(\theta), \sigma_{\text{mix}}^2\big)$, so that the likelihood penalises mixing patterns that depart from the external matrix. Unlike the epidemiological targets, the standard deviation σ_{mix} is not fixed but estimated during calibration under a uniform prior on $[5, 20]$. This avoids having to specify *a priori* how closely the model matrix should reproduce a synthetic matrix of unknown accuracy for this setting: the strength of the penalty is instead determined by the extent to which the epidemiological data and the external matrix can be reconciled. The three mixing parameters (background level, assortativity spread, and parent–child strength) are therefore informed jointly by this penalty and by the epidemiological targets.

9.3 Metropolis Algorithm

We perform calibration using Markov chain Monte Carlo (MCMC) sampling with the `DEMetropolisZ` algorithm implemented in `PyMC`. We run eight parallel chains to improve exploration of the posterior distribution. Each chain uses 10,000 tuning (adaptation) iterations, followed by 20,000 posterior draws, giving 240,000 model evaluations per calibration. For the subsequent scenario projections, we discard an initial burn-in period of 10,000 draws and retain 2,000 posterior samples per calibrated run.

Convergence is assessed by inspecting trace plots, posterior distribution smoothness, and the Gelman-Rubin diagnostic ($\hat{R}$) being below 1.05. Posterior samples from the calibrated model are propagated forward to generate projections under each intervention scenario, thereby fully accounting for parameter uncertainty.

Each model configuration is calibrated independently, and the configurations are run in parallel as array jobs on a high-performance computing cluster, each using eight cores. The full analysis required between approximately 10 and 15 hours, depending on the parameter values explored.

10 Parameter Definitions, Values and Priors

In this section, we describe the sources and rationale underlying the parameter values and prior distributions used in the model. The complete list of parameter definitions, numerical values, calibration priors, and references is provided in Table 1. Parameters were informed using a combination of published literature, previous modelling studies, empirical observations from Kiribati, and calibration to epidemiological data. Where direct evidence was unavailable or substantial uncertainty existed, prior distributions were specified to reflect plausible ranges and updated through Bayesian calibration.

10.1 TB natural history parameters

The model incorporates a detailed representation of the spectrum of *M. tuberculosis* infection and disease. Parameters governing progression through early infection states were informed by our previous work on TB latency parameterisation, which estimated parameters for different model structures using TB progression data [REFERENCE PLACEHOLDER]. While the model structure used in this analysis differs from those previously calibrated, we can establish a correspondence between key parameters based on their role within the early infection process.

The previous parameterisation study calibrated several alternative latent infection structures. The comparison presented here focuses on Model 4, which represented early infection using two latent compartments. Individuals entered an initial latent state and could either progress directly to active TB or transition to a secondary latent state associated with slower progression dynamics. The parameter ϵ governed direct progression from the initial latent state to disease, while the parameter κ governed transition from the initial latent state into the secondary latent state.

In the present model, newly infected individuals enter an incipient infection state (E), from which they may either progress to active disease or transition to a contained infection state (C). Contained infection may subsequently undergo immune-mediated clearance to X or return to the incipient state following immunological breakdown.

The progression parameter in the present model, denoted ϵ_i, governs direct progression from the initially occupied infection state (E) to active disease. This parameter therefore has the same interpretation as ϵ in Model 4, namely the rate at which newly infected individuals progress directly to disease before entering a more controlled infection state. This correspondence can be demonstrated analytically by considering disease incidence immediately following infection. When all individuals are initialised in the high-risk infection state, the incidence rate at $t = 0$ reduces to the direct progression parameter in both model structures, establishing the equivalence of ϵ and ϵ_i with respect to early progression dynamics. Consequently, estimates of ϵ from the previous calibration were used directly to inform age-specific values of ϵ_i.

No direct mathematical correspondence exists between the containment rate in the present model (γ_i) and the transition rate κ estimated in Model 4. The present formulation includes additional pathways, notably immune breakdown from the contained state back to the incipient state, which were not represented in the earlier model. Nevertheless, κ provides the most relevant source of information because both parameters govern movement from an initially infected state with elevated risk of disease progression into a state associated with greater immune control and lower short-term progression risk. Estimates of κ were therefore used to inform γ_i in the present model.

Importantly, the revised model allows the balance between rapid progression and longer-term latent infection to be determined jointly by the containment rate (γ_i) and the immune breakdown rate (ω). Calibration of the breakdown parameter therefore provides flexibility to reproduce a range of early infection trajectories that could not be represented within the simpler Model 4 structure.

The remaining parameter governing early infection dynamics, the clearance rate (ξ), describes spontaneous resolution of contained infection and has no direct analogue in the previous model. This parameter was assigned a prior distribution reflecting the plausible range of spontaneous clearance rates reported in the literature [REFERENCE PLACEHOLDER] and updated through calibration.

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Similarly, parameters governing susceptibility to reinfection following containment, infection clearance, or recovery from disease were assigned prior distributions reflecting current uncertainty in the literature and were updated through calibration.

The active disease component distinguishes individuals according to both clinical status (subclinical versus clinical disease) and infectiousness (less infectious versus more infectious disease). Parameters governing transitions between these disease states were selected to reproduce realistic distributions of disease phenotypes while remaining identifiable from the available data. To reduce identifiability issues, transition rates from more infectious to less infectious disease and from clinical to subclinical disease were fixed at values corresponding to an average duration of approximately one year in each state before transition. Transition rates in the opposite direction were estimated during calibration.

Relative infectiousness parameters were informed by the emerging literature on subclinical TB and heterogeneity in transmission potential [REFERENCE PLACEHOLDER]. Individuals classified as less infectious were assumed to generate 40% of the transmission produced by more infectious individuals, while subclinical disease was assumed to be half as infectious as clinical disease. The combined infectiousness of each disease category was determined by multiplying the corresponding relative infectiousness factors.

Parameters governing TB mortality, self-recovery, and treatment relapse were informed by published estimates and previous modelling studies [REFERENCE PLACEHOLDER]. Where substantial uncertainty remained, prior distributions were assigned and updated through calibration.

The model additionally includes a transmission-scaling parameter that allows a continuum between frequency-dependent and density-dependent transmission. This parameter was introduced to account for uncertainty regarding the relationship between population density and transmission intensity in Kiribati, where increasing population density and limited habitable land may plausibly influence contact patterns and transmission risk. The parameter was assigned a uniform prior spanning the full range between purely frequency-dependent and purely density-dependent transmission.

10.2 Programmatic parameters

Table 1: Model parameters

`Parameter`	**Definition**	`Value / Prior`	**Unit**
`raw_transmission_rate`	Effective rate of transmission (before adjusting for infectiousness)	`uniform (0.0001, 0.003)`	
`bg_mixing`	Background age-agnostic mixing level	`uniform (0.01, 0.05)`	
`a_spread`	Spread of assortative mixing pattern (smaller value means more assortativity)	`uniform (2.0, 15.0)`	
`pc_strength`	Strength of parent-children mixing pattern	`uniform (0.01, 1.0)`	
`rel_sus_contained`	Rel. risk of reinfection from 'contained' compartment (ref. 'mtb-naïve')	`uniform (0.2, 0.5)`	
`rel_sus_cleared`	Rel. risk of reinfection from 'cleared' compartment (ref. 'mtb-naïve')	`uniform (0.5, 1.0)`	
`rel_sus_children`	Rel. susceptibility to infection for under 15 year-old individuals (ref. 15 and over)	`uniform (0.3, 1.0)`	
`rel_infectiousness_subclin`	Rel. infectiousness of subclinical TB (ref. clinical TB)	`0.5`	
`rel_infectiousness_lowinf`	Rel. infectiousness of 'less infectious' TB (ref. 'more infectious' TB)	`0.4`	
`progression_rate_age0`	Rate of progression from 'incipient' to TB disease (age 0-4)	`2.4`	/year

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Table 1: Model parameters (continued)

Parameter	Definition	Value / Prior	Unit
progression_rate_age5	Rate of progression from 'incipient' to TB disease (age 5-14)	2.0	/year
progression_rate_age15	Rate of progression from 'incipient' to TB disease (age 15 and over)	0.1	/year
progression_prop_infectious	Proportion of incident TB that is 'more infectious'	0.5	/year
containment_rate_age0	Rate of transition from 'incipient' to 'contained' (age 0-4)	4.4	/year
containment_rate_age5	Rate of transition from 'incipient' to 'contained' (age 5-14)	4.4	/year
containment_rate_age15	Rate of transition from 'incipient' to 'contained' (age 15 and over)	2.0	/year
breakdown_rate	Rate of transition from 'contained' to 'incipient' (all ages)	uniform (0.01, 1.0)	/year
clearance_rate	Rate of transition from 'contained' to 'cleared' (all ages)	uniform (0.01, 0.1)	/year
clinical_progression_rate	Rate of progression from subclinical to clinical TB	uniform (0.5, 5.0)	/year
clinical_regression_rate	Rate of transition from clinical to subclinical TB	1.0	/year
infectiousness_gain_rate	Rate of progression from 'less infectious' to 'more infectious' TB	uniform (0.5, 10.0)	/year
infectiousness_loss_rate	Rate of transition from 'more infectious' to 'less infectious' TB	1.0	/year
tb_mortality_rate_inf	Rate of TB mortality for 'more infectious' clinical disease	0.389	/year
tb_mortality_rate_lowinf	Rate of TB mortality for 'less infectious' clinical disease	0.025	/year
self_recovery_rate	Rate of self-recovery for subclinical TB	0.4	/year
recent_detection_rate	Annual rate of TB detection in 2020	uniform (0.1, 5.0)	/year
tx_duration	Average TB treatment duration	0.5	year
pct_neg_tx_death	Percentage of deaths among negative TB treatment outcomes	40.0	%
prev_se_incipient_tst	Probability of TST positivity for 'incipient' state	0.75	
prev_se_contained_tst	Probability of TST positivity for 'contained' state	0.75	
prev_se_cleared_tst	Probability of TST positivity for 'cleared' and 'recovered' states	uniform (0.2, 0.5)	
prev_se_subclin_lowinf_pearl	Probability PEARL-positive for 'subclinical' and 'less infectious' category	1.0	
prev_se_clin_lowinf_pearl	Probability PEARL-positive for 'clinical' and 'less infectious' category	1.0	
prev_se_subclin_inf_pearl	Probability PEARL-positive for 'subclinical' and 'more infectious' category	1.0	
prev_se_clin_inf_pearl	Probability PEARL-positive for 'clinical' and 'more infectious' category	1.0	
prev_se_subclin_lowinf_plts	Probability PLTS-positive for 'subclinical' and 'less infectious' category	0.3	

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Table 1: Model parameters (continued)

Parameter	Definition	Value / Prior	Unit
prev_se_clin_lowinf_plts	Probability PLTS-positive for 'clinical' and 'less infectious' category	1.0	
prev_se_subclin_inf_plts	Probability PLTS-positive for 'subclinical' and 'more infectious' category	0.5	
prev_se_clin_inf_plts	Probability PLTS-positive for 'clinical' and 'more infectious' category	1.0	
prev_se_subclin_lowinf_cxr	Probability CXR-positive for 'subclinical' and 'less infectious' category	uniform (0.07, 0.52)	
prev_se_clin_lowinf_cxr	Probability CXR-positive for 'clinical' and 'less infectious' category	1.0	
prev_se_subclin_inf_cxr	Probability CXR-positive for 'subclinical' and 'more infectious' category	uniform (0.72, 0.93)	
prev_se_clin_inf_cxr	Probability CXR-positive for 'clinical' and 'more infectious' category	1.0	
prev_se_subclin_lowinf_ssx	Probability SSx-positive for 'subclinical' and 'less infectious' category	0.0	
prev_se_clin_lowinf_ssx	Probability SSx-positive for 'clinical' and 'less infectious' category	1.0	
prev_se_subclin_inf_ssx	Probability SSx-positive for 'subclinical' and 'more infectious' category	0.0	
prev_se_clin_inf_ssx	Probability SSx-positive for 'clinical' and 'more infectious' category	1.0	
tpt_completion_perc	TPT completion percentage	70.0	%
rel_detection_subclin	Relative detection rate of subclinical TB under passive case finding (ref. clinical TB)	0.0	
passive_detection_inflection	Time when passive detection started to scale up	uniform (1990.0, 2020.0)	
passive_detection_shape	Shape parameter of passive detection scale-up profile	0.15	
passive_detection_past_frac	Past passive detection rate, as a fraction of the current one	uniform (0.3, 1.0)	
infection_pop_scale	Exponent for population scaling in force of infection calculation	uniform (0.0, 1.0)	
reachable_pop_frac	Proportion of population that is reachable by screening interventions	0.85	
rel_detection_unreachable	Relative detection rate of unreachable population under passive case finding (ref. reachable)	0.5	
rel_sus_unreachable	Relative susceptibility to infection of unreachable population (ref. reachable)	1.5	